Section A

Answer **all** the questions in the spaces provided.

1 (a) Fig. 1.1 shows the variation of force F with the extension x of a spring.

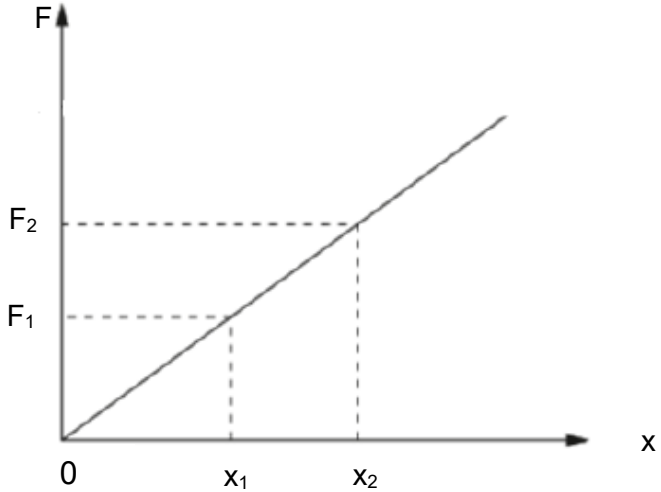

Fig. 1.1

The extension of the spring is increased from x_1 to x_2.
Show that the work done W in extending the spring is given by

$$W = \frac{1}{2} k (x_2^2 - x_1^2)$$

where k is the spring constant.

[2]

(b) Fig. 1.2 shows a signboard suspended by two elastic springs.

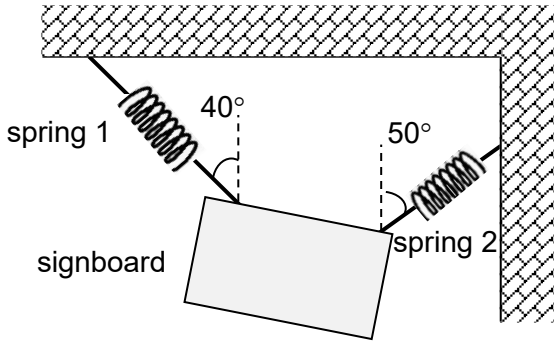

Fig. 1.2

(i) State the conditions for equilibrium.

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…….……..... [2]

(ii) On Fig. 1.2, draw and label clearly the forces acting on the signboard. Mark the centre of gravity of the signboard with a dot and label the point as G. [2]

(iii) Given that the tension in spring 1 is 300 N and the tension in spring 2 is 252 N, determine the weight of the signboard.

weight = …….. N [2]

(iv) The signboard is pulled vertically downwards with a force of 20 N so that the springs are stretched to a new position. Determine the acceleration of the signboard immediately after it is released.

acceleration = …….m s^{-2} [2]